THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 3: **The World of Metals and Non-metals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The solution of magnesium oxide in water turns red litmus: 1

(a) colourless **(b)** blue **(c)** green **(d)** red

2. Which non-metal is a good conductor of electricity? 1

(a) sulphur **(b)** graphite **(c)** iodine **(d)** phosphorus

3. Aluminium foil is used for wrapping food because it is: 1

(a) a non-metal **(b)** dull **(c)** malleable **(d)** brittle

4. Metals react with dilute acids to produce: 1

(a) hydrogen gas **(b)** carbon dioxide **(c)** oxygen gas **(d)** nitrogen

5. The hardest natural substance is: 1

(a) sulphur **(b)** diamond **(c)** graphite **(d)** iron

6. Cooking vessels are made of metals because they are: 1

(a) poor conductors **(b)** transparent
(c) good conductors of heat **(d)** brittle

7. The reaction $\mathrm{Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2}$ produces: 1

(a) zinc carbonate **(b)** zinc oxide and water
(c) zinc chloride and hydrogen **(d)** zinc and chlorine

8. Which of these metals is soft enough to be cut with a knife? 1

(a) sodium **(b)** copper **(c)** zinc **(d)** iron

9. Bells are made of metal because metals are: 1

(a) sonorous **(b)** soft **(c)** brittle **(d)** dull

10. The gas evolved when a metal reacts with an acid burns with a: 1

(a) green flame **(b)** smoke **(c)** no sound **(d)** pop sound

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Give two uses of aluminium. 2

12. Why are cooking vessels made of metals? 2

13. Write the reaction of zinc with dilute hydrochloric acid. 2

14. Why are gold and silver used for jewellery? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe the effect of burning magnesium and sulphur in oxygen and testing the products with litmus. 3

16. Explain sonority with an example. 3

17. How can you distinguish a metal from a non-metal in a sample? 3

OR

What happens when iron nail is dipped in copper sulphate solution? Write the equation. 3

18. Why are metals used for making machines and bridges? 3

OR

Mention any three uses of metals in daily life. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe an activity to show displacement of copper from copper sulphate by iron and zinc. 5

OR

Explain the importance of metals in the development of civilisation. 5

20. Explain the uses of metals and non-metals in daily life with five examples. 5

OR

Explain the formation of metal oxides and non-metal oxides with two examples and their nature. 5

END OF QUESTION PAPER — Total: 40 Marks

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